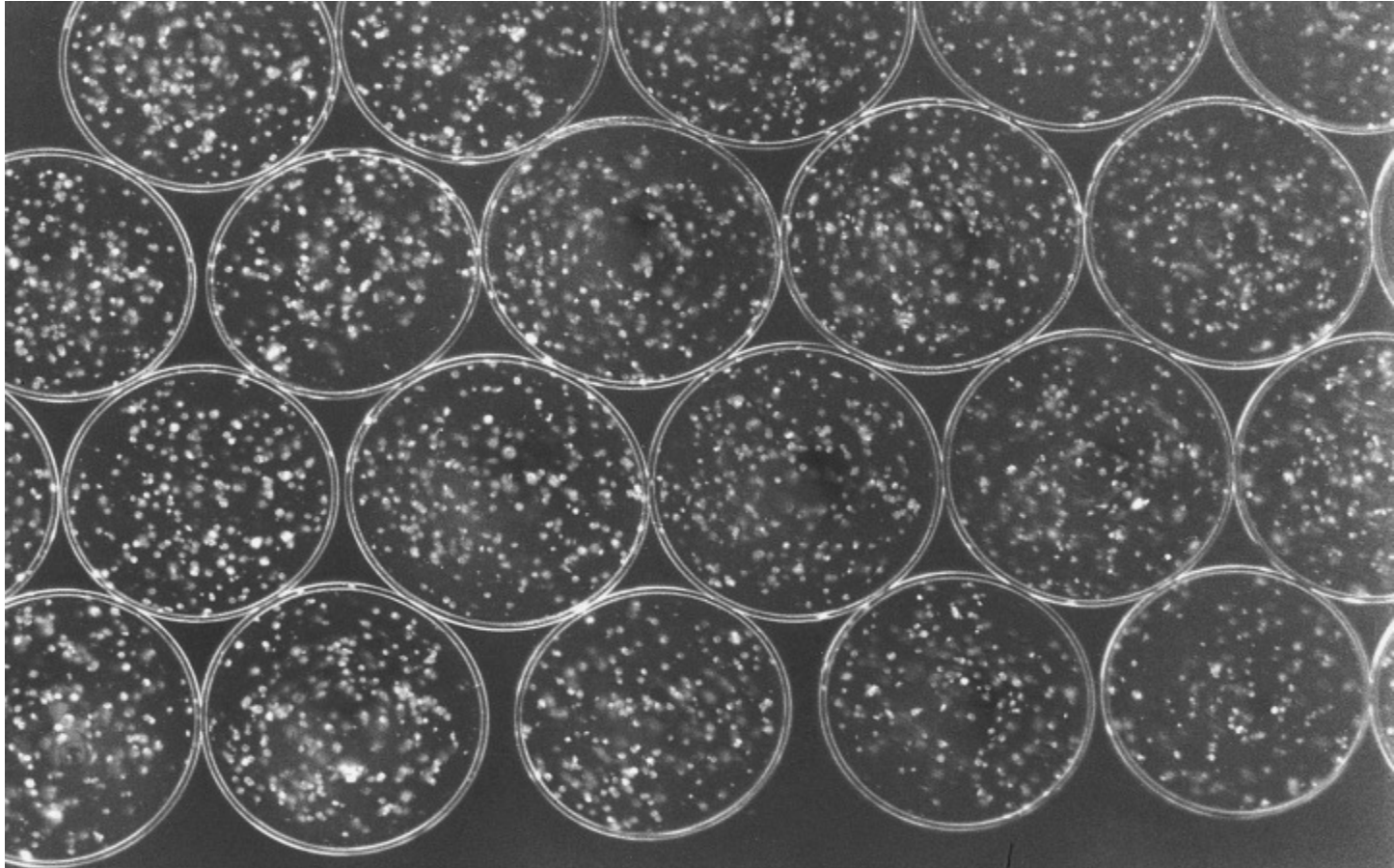


Gene Therapy

ENCAPSULATED MYOBLASTS



Implantation in immuno-deficient mouse

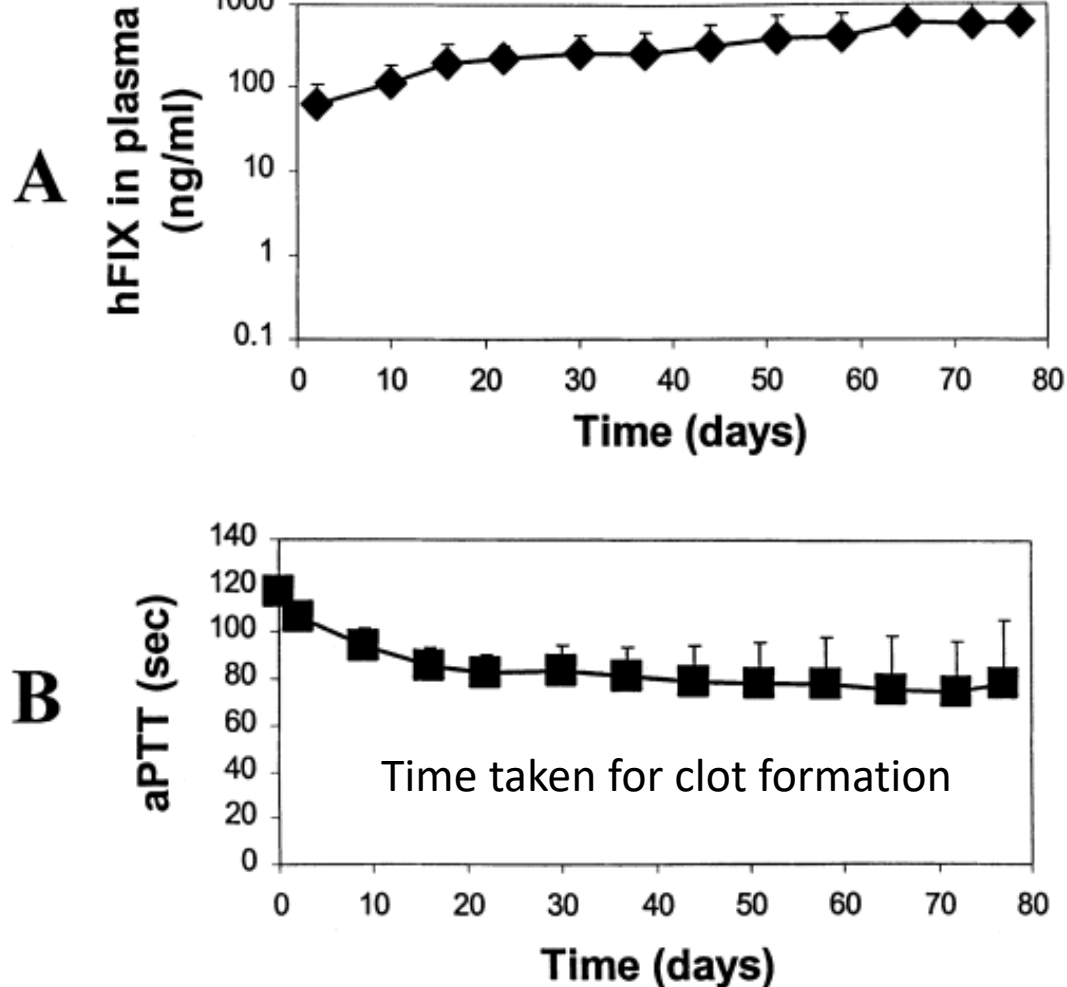
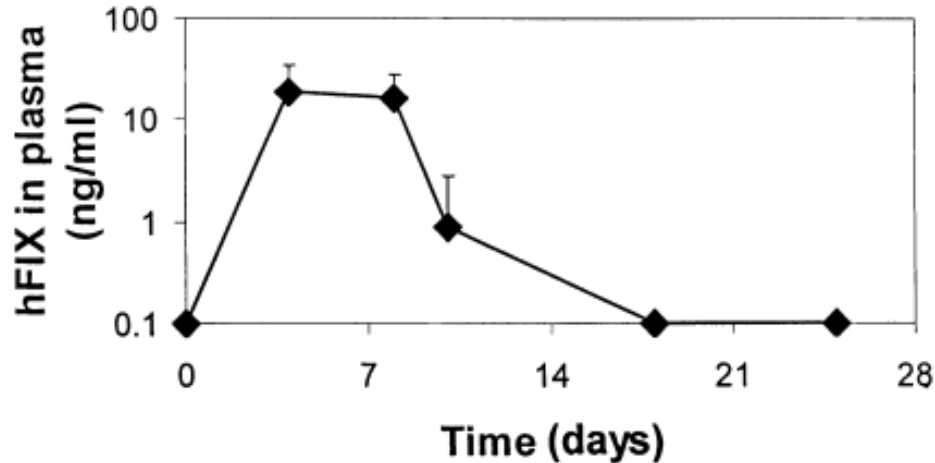


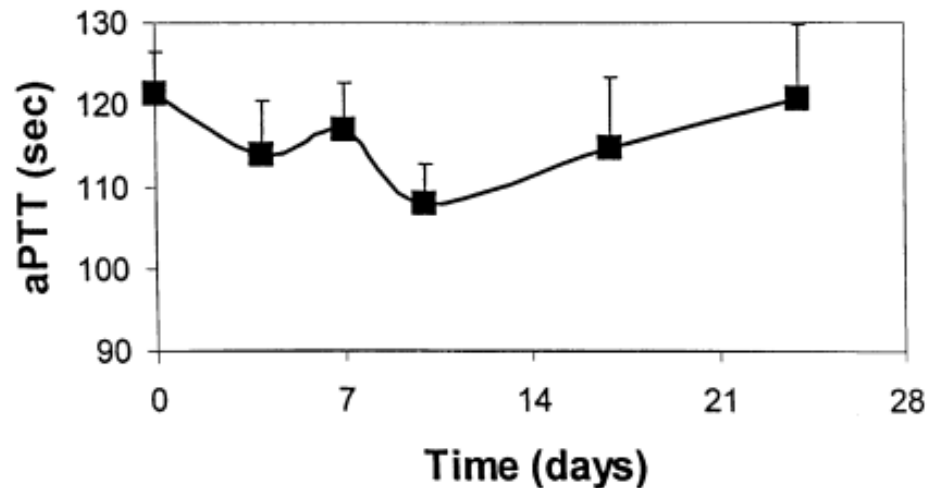
Fig . FIX and APTT in nude haemophilic mice. Determination of hFIX antigen in the circulation of nude haemophilic mice. Haemophilic immunodeficient mice ($n=7$) were implanted with microencapsulated mMyohFIX-18 myoblasts. At various times postimplantation, mice blood samples were collected through retroorbital bleeding. **(A)** hFIX antigen levels in plasma were assayed by ELISA. **(B)** Clotting activity was determined by APTT. Error bars are the standard deviations. Antigen hFIX levels were sustained in treated mice for at least 77 days (length of experiment).

Implantation in immuno-competent mouse

A



B



Factor IX and APTT in immuno-competent haemophilic mice. Delivery of hFIX in immuno-competent haemophilia mice implanted with encapsulated myoblasts transfected with pLNMIXIL. Haemophilic mice (n=7) were implanted with microencapsulated myoblasts. At various times postimplantation, blood samples were collected from each mouse through retroorbital bleeding. (A) hFIX antigen levels in plasma were assayed by ELISA. (B) The functional activity of the hFIX delivered was determined by the APTT of the plasmas obtained from the implanted mice.

Human protein in mouse: Antibodies

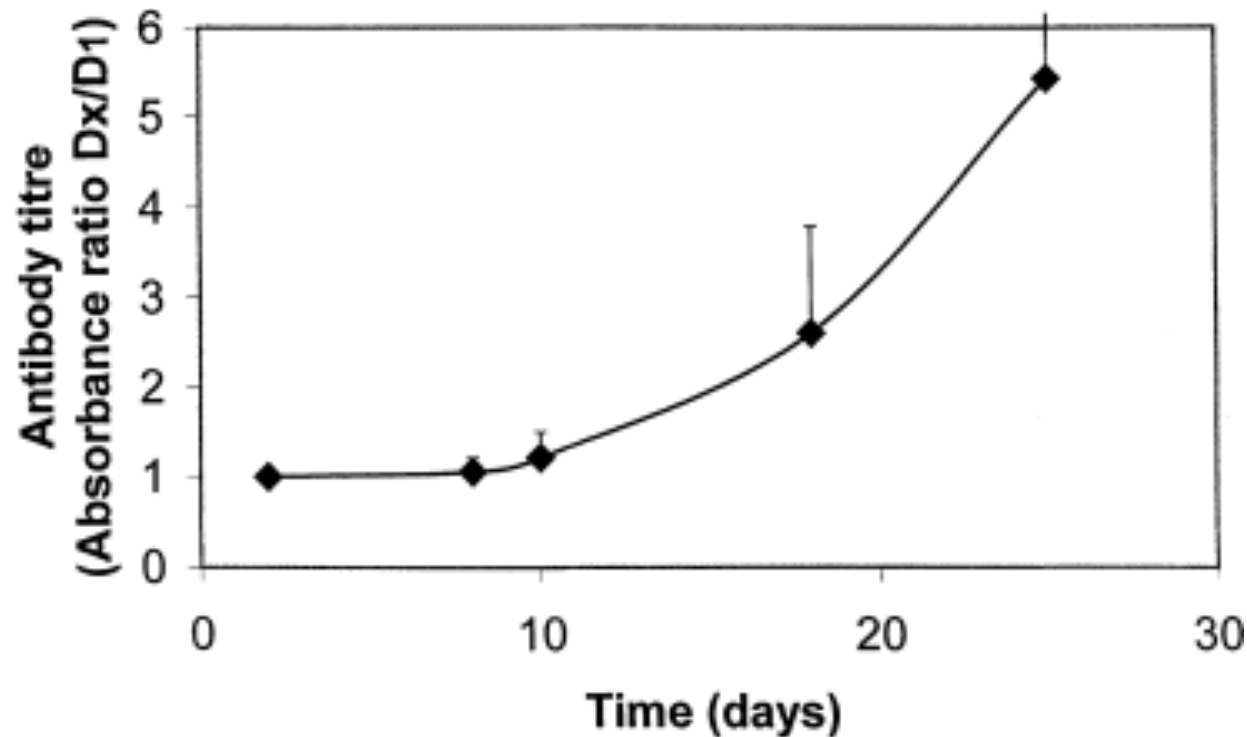


Fig. Antibodies to hFIX in immunocompetent mice implanted with microcapsules. AntihFIX antibodies in the plasmas of immunocompetent mice implanted with encapsulated myoblasts. The presence of antibodies to hFIX in mouse plasma was detected by ELISA, using mouse plasma diluted 1 : 100. The ratio of absorbance (OD at 405 nm) obtained for a given blood sample (Dx) over that of the blood sample from the same mouse at day 0 (D0), was plotted to measure the antibody titre. Each value shows the average antibody titre in the plasmas of each group of mice. Error bars are the standard deviation.

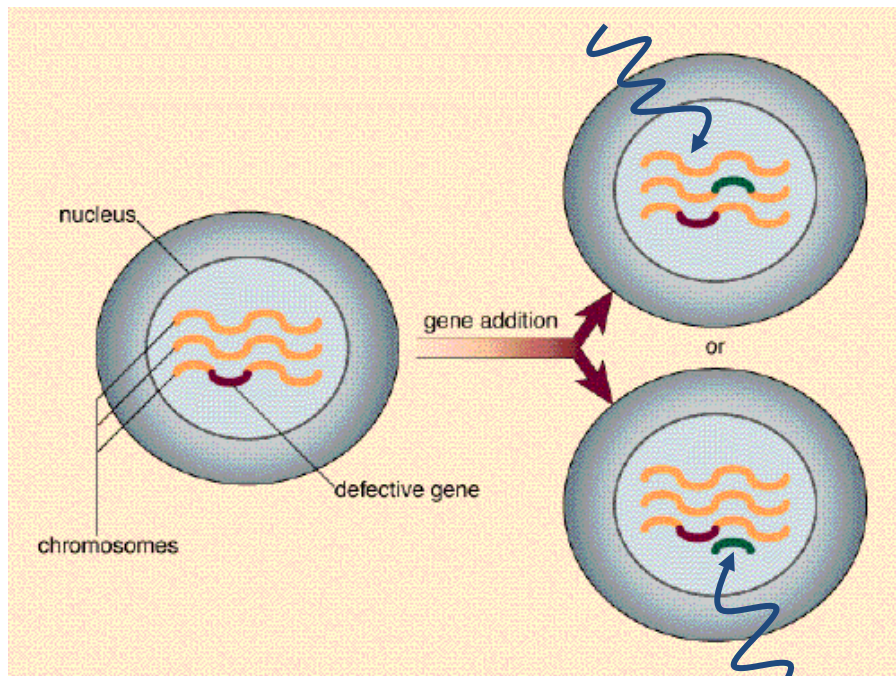
What is Gene Therapy (no cell delivery)?

Gene Therapy means treating a disease by introducing some genetic material (nucleic acid) into our cells. These genes can:

- a. Replace a missing or defective gene
- b. Repair a non-functional or mal-functional gene
- c. Produce a therapeutic protein that helps to treat the disease
- d. Can vaccinate the patient against an infection, etc.

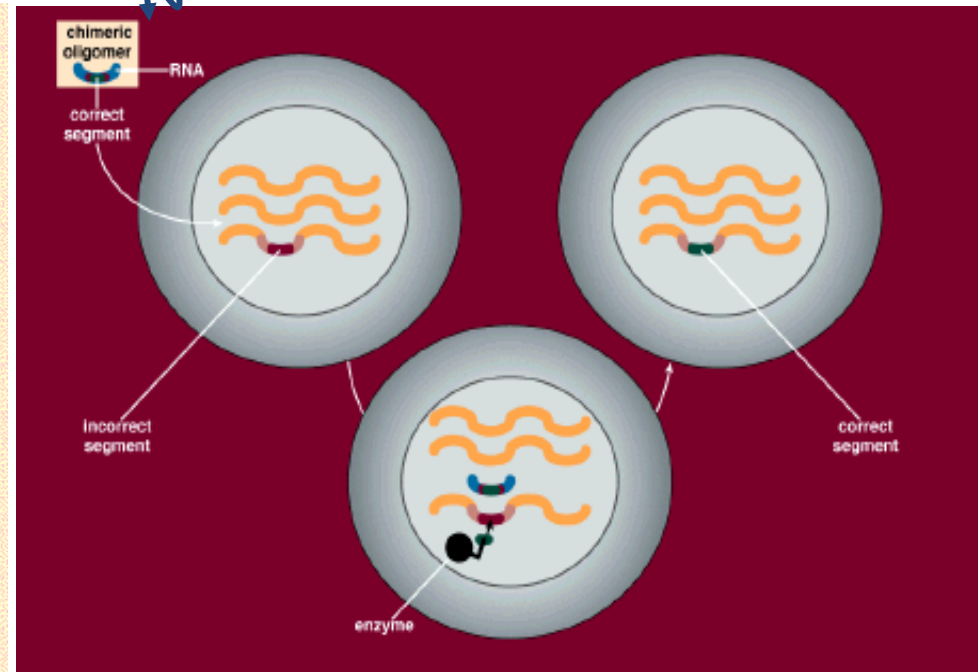
What is Gene Therapy?

Chromosomal insertion



Gene Addition or Replacement

Extra-chromosomal



Gene Repair

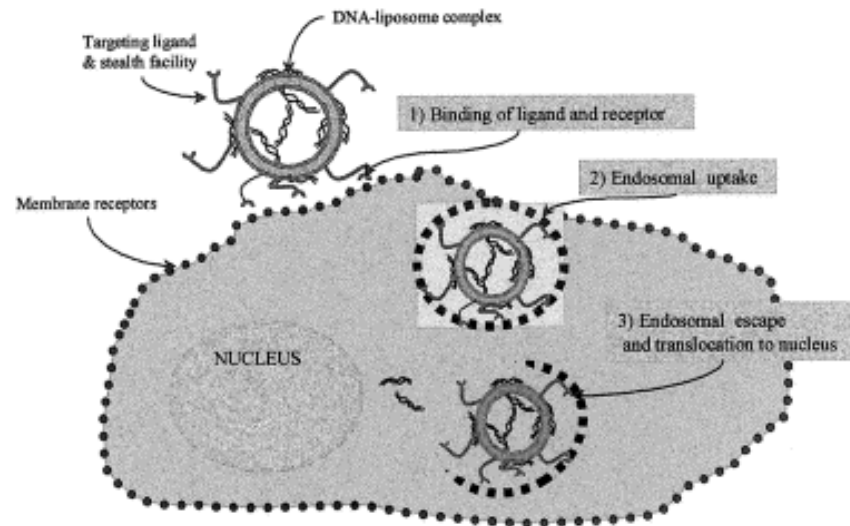
From: Eric B. Kmiec, American Scientist, May-June 1999

Barriers in Gene Delivery



EXTRACELLULAR BARRIERS

- Degradation of DNA in plasma
- Uptake of DNA by reticuloendothelial system
- Inability to target DNA to specific organs
- Largely ineffective via the oral route – except for immunisation
- Transfection inhibited by mucus



INTRACELLULAR BARRIERS

- Endosomal escape of DNA
- Lysosomal degradation of DNA
- Cytoplasmic stability of DNA
- Translocation of DNA to the nucleus

Understanding the barriers: Intracellular studies

$$x_{rms} = \sqrt{2Dt}$$

Efficient active transport of gene nanocarriers to the cell nucleus

3878–3882 | PNAS | April 1, 2003 | vol. 100 | no. 7

Junghae Suh*, Denis Wirtz^{†‡§}, and Justin Hanes^{*†§}

Departments of *Biomedical Engineering, [†]Chemical and Biomolecular Engineering, and [‡]Materials Science and Engineering, Molecular Biophysics Program, The Johns Hopkins University, 3400 North Charles Street, Baltimore, MD 21218

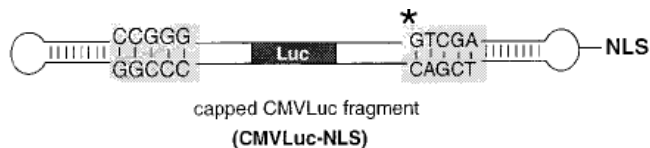


FIG. 1. Strategy for the preparation of a double-stranded DNA fragment coupled to an NLS peptide. A functional luciferase gene of 3,380 bp was cut out of pCMVLuc with *Xma*I and *Sa*II. Further digestion with *Xmn*I and *Bsp*HI cut the unwanted restriction fragment into small fragments (970, 875, 768, and 240 bp) that were removed by sucrose gradient centrifugation. The capped CMVLuc-NLS DNA was obtained by ligation of the ³²P-labeled (*) oligonucleotide-peptide and oligonucleotide-cap hairpins to the restriction fragment.

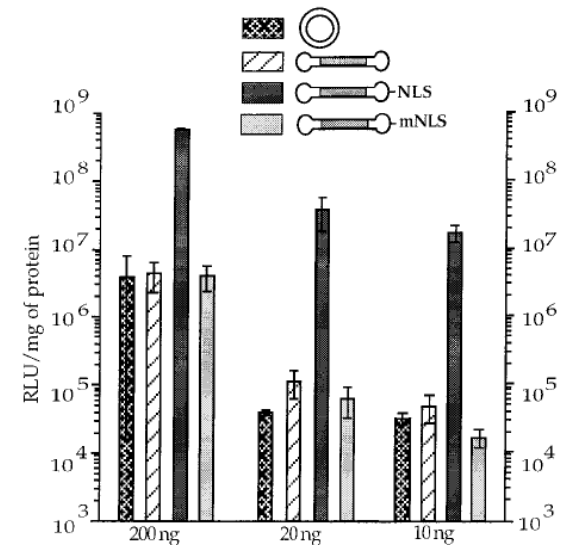
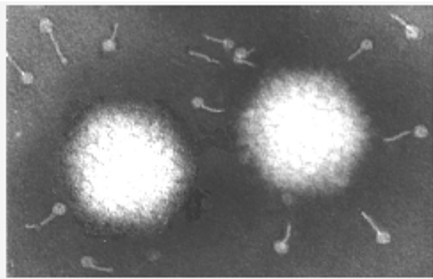
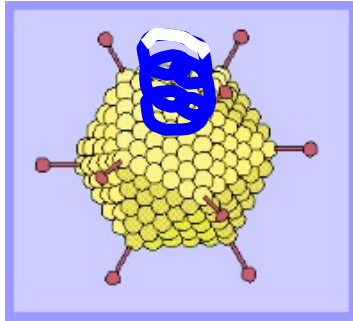


FIG. 5. Sustained luciferase expression levels are due to the nuclear localization peptide. HeLa cells were transfected with various amounts of DNA complexed to ExGen500 PEI (N/P = 5 in a 5% glucose solution) in the presence of 10% serum.

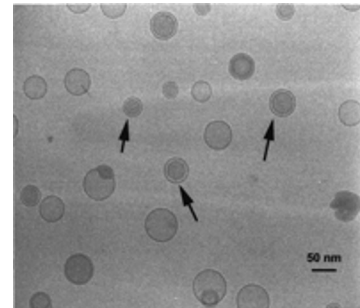
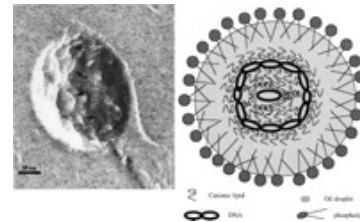
Carriers for putting genes inside cells?

We need a carrier

Virus



Polymers and Lipids (Non-viral)



The State of the Science: Viral & Non-viral

Viral Vectors

Non-viral Vectors

Pros

- 1. High Transfection Efficiency**
- 2. Natural Tropism (ability to infect different cells)**
- 3. Evolved mechanisms for endosomal escape**
- 4. Natural transportation mechanism of DNA into nucleus**

Cons

- 1. Strong immune reactions against viral proteins prohibit multiple administrations**
- 2. Possibility of chromosomal insertion and proto-oncogene activation**
- 3. Complicated synthesis process**
- 4. Limitation on gene size**
- 5. Toxicity, contamination of live virus**

- 1. Low immunogenicity**
- 2. Can be made to be non-toxic**
- 3. Easy to synthesize: quality control for mass production**
- 4. Potentially targetable**
- 5. No limit on plasmid size**
- 6. No integration: Can be administered as drugs**

- 1. Low transfection efficacy**
- 2. Currently most vehicles are toxic at high doses**
- 3. No natural tropism, endosomal escape or nuclear transport mechanisms**

Betibeglogene autotemcel

- Beta thalassemia is caused by mutations to or deletions of the HBB gene leading to reduced or absent synthesis of the beta chains of hemoglobin that result in variable outcomes ranging from severe anemia to clinically asymptomatic individuals.
- LentiGlobin BB305 is a lentiviral vector which inserts a functioning version of the HBB gene into a recipient's blood-producing hematopoietic stem cells (HSC) **ex vivo**.
- The resulting engineered HSCs are then reintroduced to the recipient.
- Approved for medical use in the European Union in May 2019

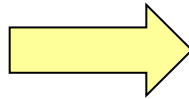
Gene Delivery: Where and How?

Macroscopic Delivery



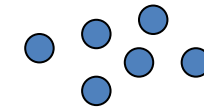
Injections, Nasal spray, Oral, Skin

Engineering novel carriers



Microscopic Delivery

Targeting,
Binding
and Uptake



Endosomal
Escape

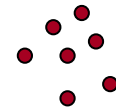


Cytoplasmic
Release



Nuclear
Transport

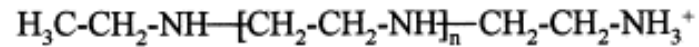
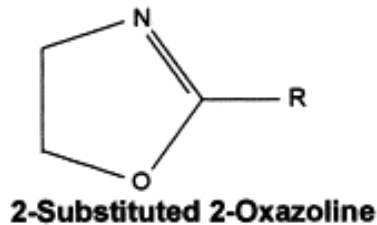
Protein
Expression



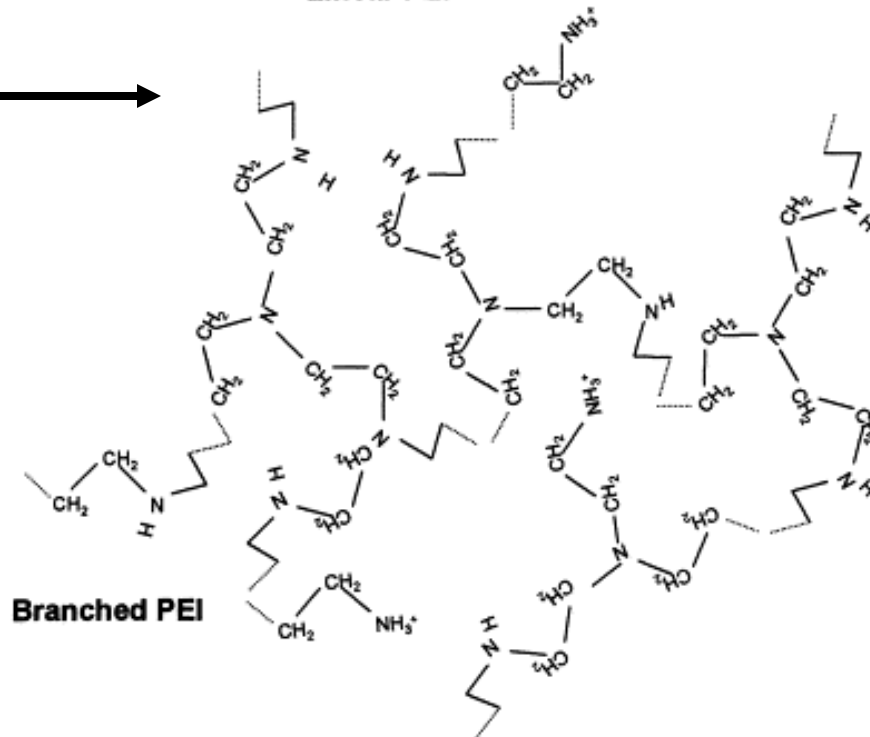
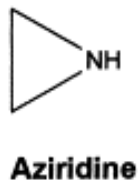
Polymers used in Gene therapy

- Cationic polymers
 - Why Cationic? DNA itself is a highly negatively charged polymer
 - Can form complexes with DNA and create “packets” or “particles”
 - One can modify them to attach targeting ligands
 - Can be “eaten” (endocytosis) by cells → transfection
 - E.g: Poly(lysine), Poly(ethyleneimine), Chitosan, Polyamidoamine dendrimers
- Other polymers
 - Can slowly release genes (DNA) from polymer devices inside the body
 - Can be in the form of small particles or matrices or gels
 - E.g: Poly(lactide-co-glycolide) or PLGA, Polyvinylpyrrolidone (PVP) etc.

Polyethyleneimine as gene carrier

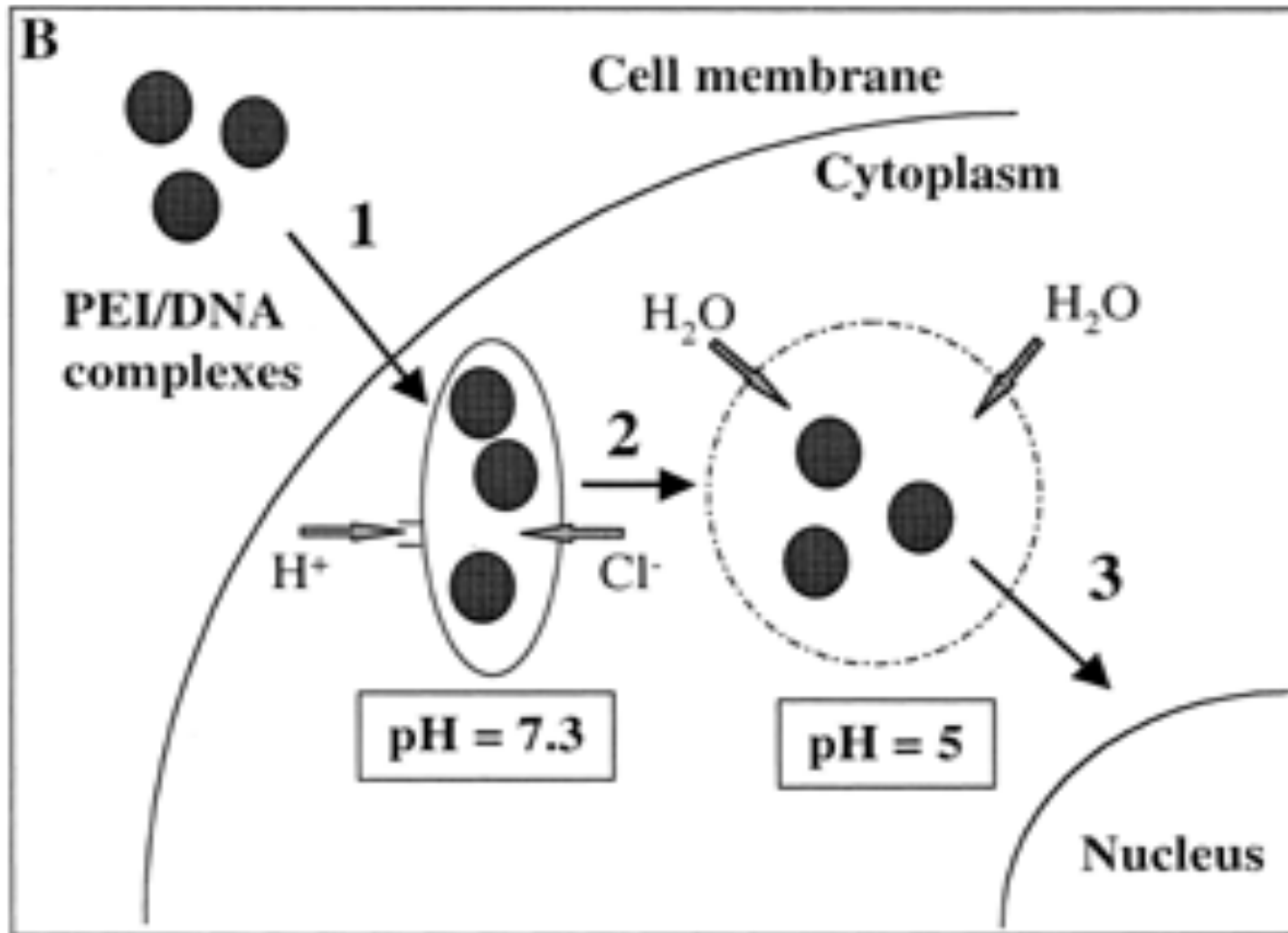


Linear PEI*



Cationic chain-growth polymerization process

Polyethyleneimine as gene carrier



PEI can be toxic and hence care must be taken before use

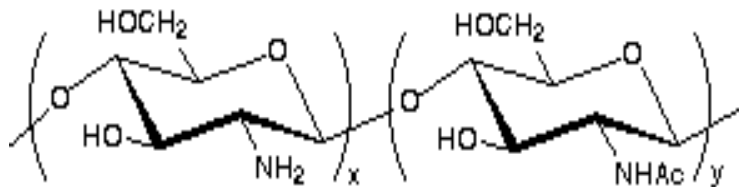
Chitosan as a gene carrier

■ Chitosan

- Cationic bio-polymer that can complex with anionic DNA
- Natural polysaccharide, **non-toxic**, non-immunogenic, biocompatible
- Suitable for mucosal delivery (mucoadhesive)

■ Complex coacervation

- Available amine groups on surface: Ligand attachment and potential for targeted delivery



General Structure of Chitosan

Vaccine

- Antigen: Molecule against which the immune response needs to be generated
 - Mostly protein based
- Adjuvant: Molecules that stimulate immune response to the target antigen
 - Amplifies the response
 - Can be used to direct the response towards a certain type: Cellular vs Humoral
 - Several different types and function

Table 1

Description of Toll-like receptor (TLR) ligands

TLR-L number	Ligand	Description (Source)
1	Pam ₃ Cys	Synthetic lipoprotein
2	LTA	Lipoteichoic acid (<i>St. aureus</i>)
	PGN	Peptidoglycan (<i>St. aureus</i> , <i>B. subtilis</i> , <i>E. coli</i>)
	HKML	Heat Killed <i>L. monocytogenes</i> (Gram+)
	LPS	Lipopolysaccharide (<i>E. coli</i>)
3	dsRNA	Oligonucleotide
	PolyI:C	Synthetic analog of dsRNA
4	LPS	Lipopolysaccharide (<i>E. coli</i>)
5	Flagellin	<i>S. typhimurium</i>
6	PGN	Peptidoglycan (<i>St. aureus</i> , <i>B. subtilis</i> , <i>E. coli</i>)
7	ssRNA	Oligonucleotide with repeating G and U motives
	Loxoribine	Guanosine analogue
	Imiquimod and Resiquimod	Small synthetic antiviral imidazoquinolines
8	Imiquimod and Resiquimod	Small synthetic antiviral imidazoquinolines
9	dsDNA	Oligodeoxynucleotide with repeating C and G motives (CpG)
10	Unknown	Unknown
11	Unknown	Unknown

Vaccine type by time of delivery

- Prophylactic vaccine
 - Given prior to the disease
 - Creates memory for the immune system
 - Widely used
 - Example: Polio
 - Rabies
 - Post exposure prophylaxis
- Therapeutic vaccine
 - Given after the disease
 - Purpose is to strengthen the immune system to fight back against the active disease
 - Still in nascent phase
 - Example: Cancer vaccines

Vaccine type by content

- Live-attenuated vaccines
 - Very strong immune response
 - Storage conditions should be stringent
 - Risky
 - Example: Smallpox, Measles
- Inactivated vaccines
 - Killed germ, example: β -propiolactone used in Covaxin
 - Lower response than live vaccines: may need several doses
 - Example: HepA, Flu, Polio
- Subunit, recombinant, polysaccharide, and conjugate vaccines
 - Can be given to anyone
 - Needs booster shots
 - Example: HPV, HepB
- Toxoid vaccines
 - Use toxin
 - Create immunity against the toxin and not the germ
 - Example: Diphtheria, Tetanus

Delivering Vaccine Antigens to APCs: Particle-based Approaches

- Microparticles: Why?
 - DCs and macrophages take up particles in large quantities
- Co-delivery of adjuvants
 - Possible to co-deliver the antigen and adjuvant for better efficacy
- Protein, peptide and DNA antigens
 - Can carry all types of antigens
- Formulation issues
 - Do suffer from effect of synthesis and loading process on antigen and adjuvant structures and activity

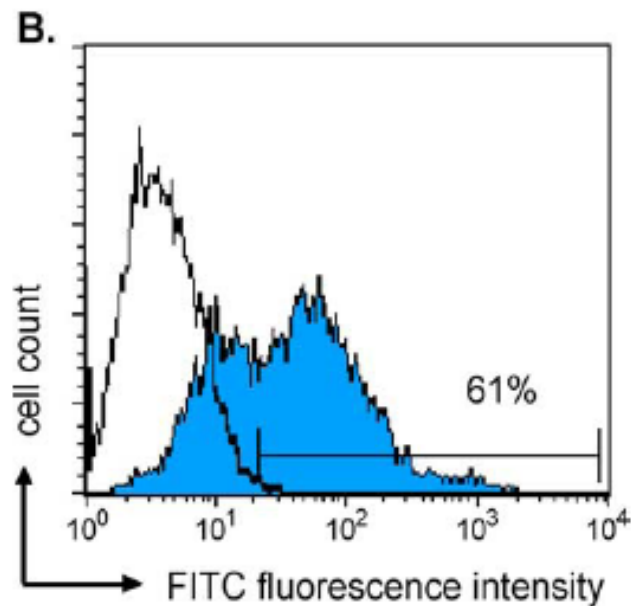
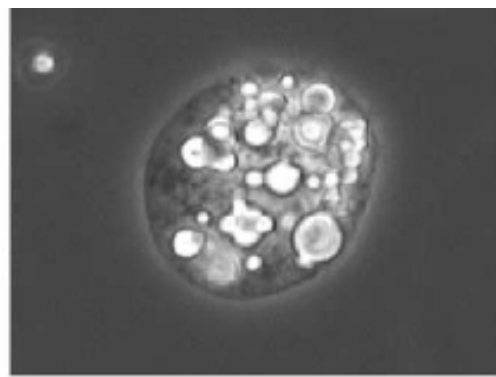
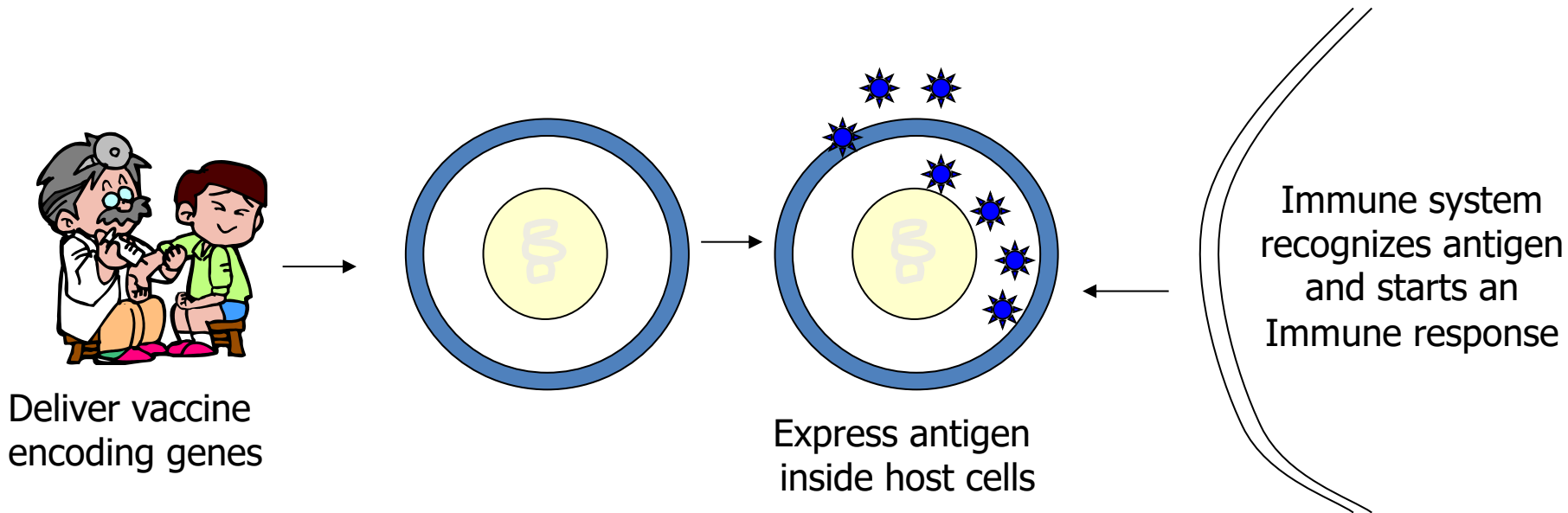


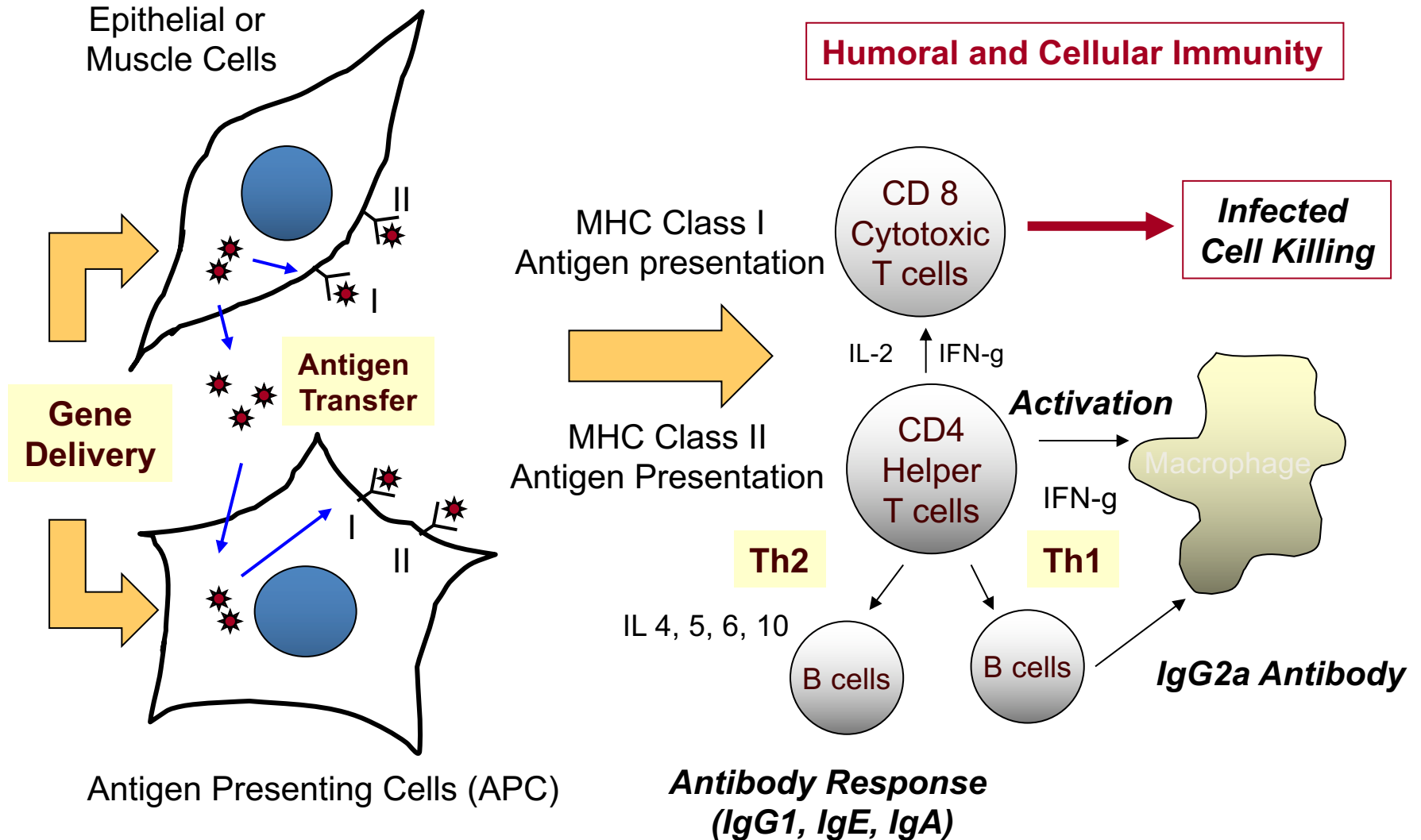
Fig. 1. Phagocytosis of fluorescent microspheres (MS) by human MoDC. Immature hMoDC were incubated with FITC containing MS for 18 h at 37 °C. Cells were then collected, washed and examined by phase contrast microscopy. (A) One immature hMoDC containing many MS is shown at magnification 40 $\times$. (B) Quantitative uptake of fluorescent MS by immature hMoDC is examined by flow cytometry. The number within the histogram plot represents the percentage of MS-FITC fluorescent hMoDC based on the whole pregated hMoDC population.

Genes as Vaccines: How?

- **Conventional vaccines:** Replication deficient bacteria, recombinant proteins (antigens) etc.
 - The antigen (immunogenic component of the pathogen) is processed by the hosts immune cells which generates a protective response
- **DNA vaccines:** Genes encoding for specific antigenic proteins of interest
 - Instead of delivering the pathogen or recombinant protein, the host cells themselves produce the antigen



DNA Vaccination: The Players



Cationic microparticles for DNA vaccines

- PLGA microparticles prepared using w/o/w or o/w emulsion techniques in the presence of a cationic surfactant
- Produces cationic surface on which plasmid DNA can be efficiently adsorbed
- Could have better “bioavailability” of the DNA compared to encapsulated plasmids
- Increases the efficacy of DNA to target Antigen presenting cells and enhances immune response

Table 1. Cationic microparticles with adsorbed DNA: Particle size, net surface charge, loading efficiency, and DNA loading levels based on hydrolysis of PLG-DNA formulation

Formulation	Mean size, μm	Zeta potential, mV	Loading level, % wt/wt	Mean loading efficiency, %
PLG/CTAB-p55 DNA	1.54 $\pm$ 0.20	+36 $\pm$ 6	0.92 $\pm$ 0.12	92.0
PLG/DDA-p55 DNA	2.20 $\pm$ 0.12	+30 $\pm$ 10	0.68 $\pm$ 0.08	68.0
PLG/DOTAP-p55 DNA	0.98 $\pm$ 0.11	+24 $\pm$ 8	0.62 $\pm$ 0.14	62.0

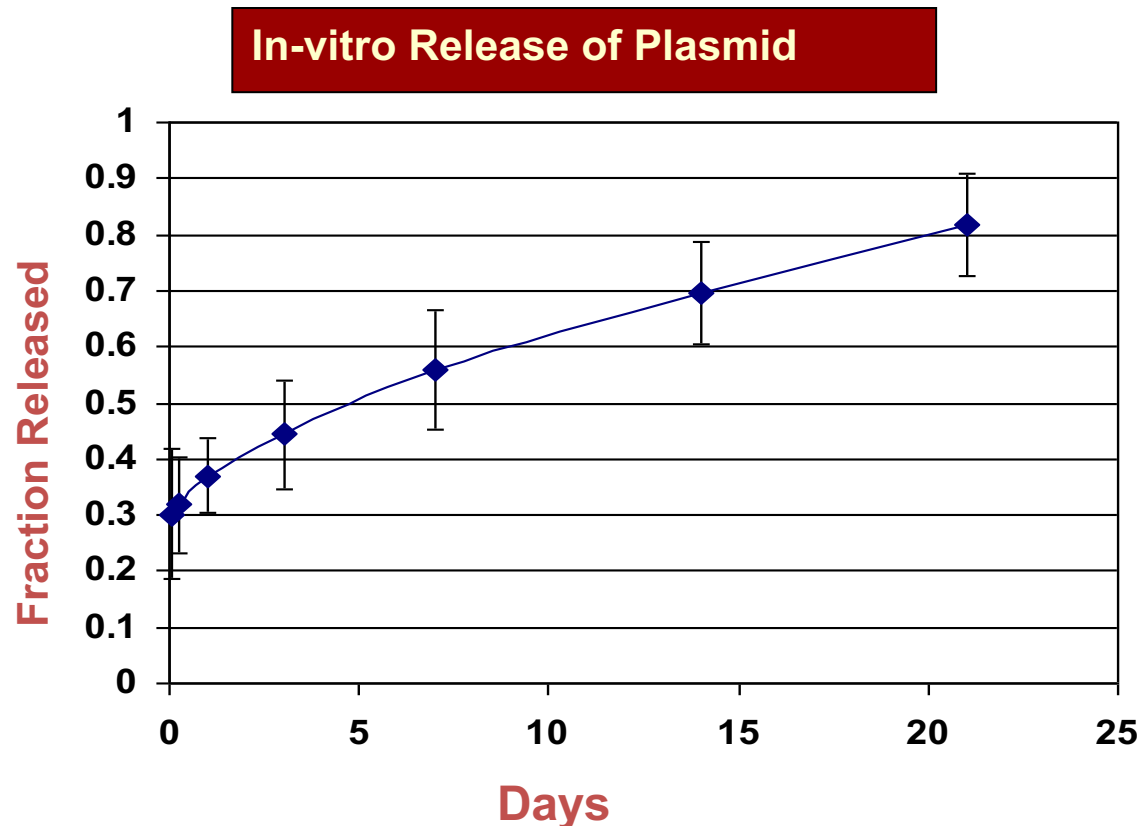
PLG microparticles prepared with polyvinyl alcohol as a particle stabilizer, but without cationic surfactants have a zeta potential of -13 ± 3 mV. Values are mean $\pm$ SE of triplicate measurements.

Developed by: Chiron Corporation (NOVARTIS), Emeryville, CA

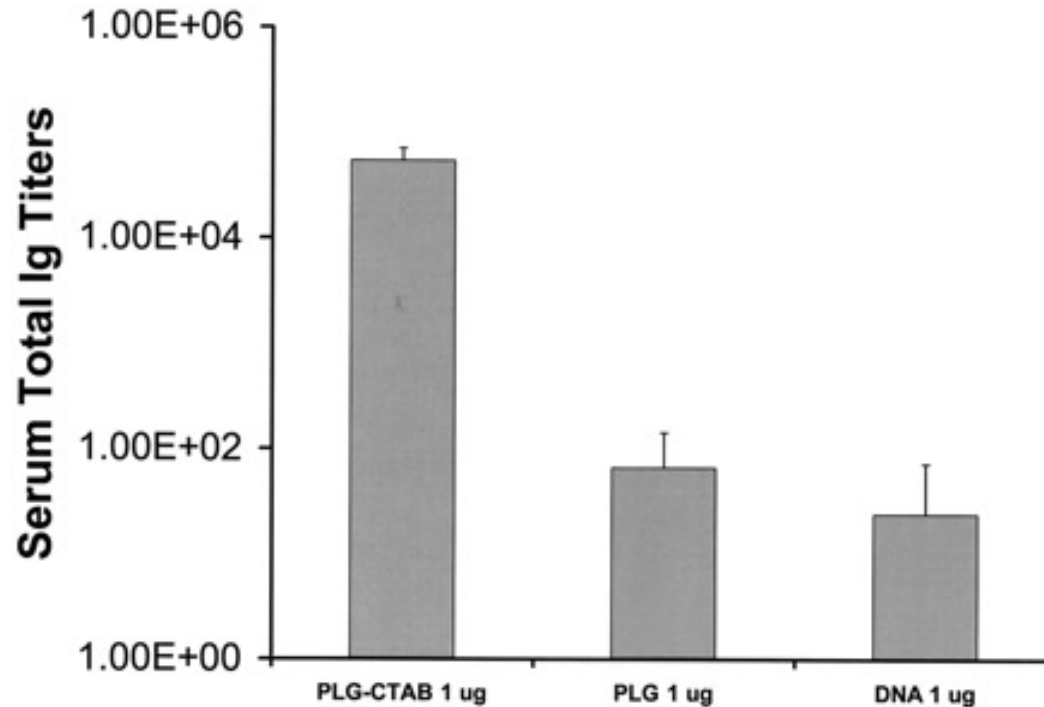
Characterization: DNA Release Profile

In-vitro release of plasmid DNA from PLGA microspheres:

Spheres were suspended in PBS/EDTA buffer and incubated at 37°C (end over end tumbling) for various time intervals. DNA concentrations were measured in the release buffer following centrifugation of the microspheres



Immune response data

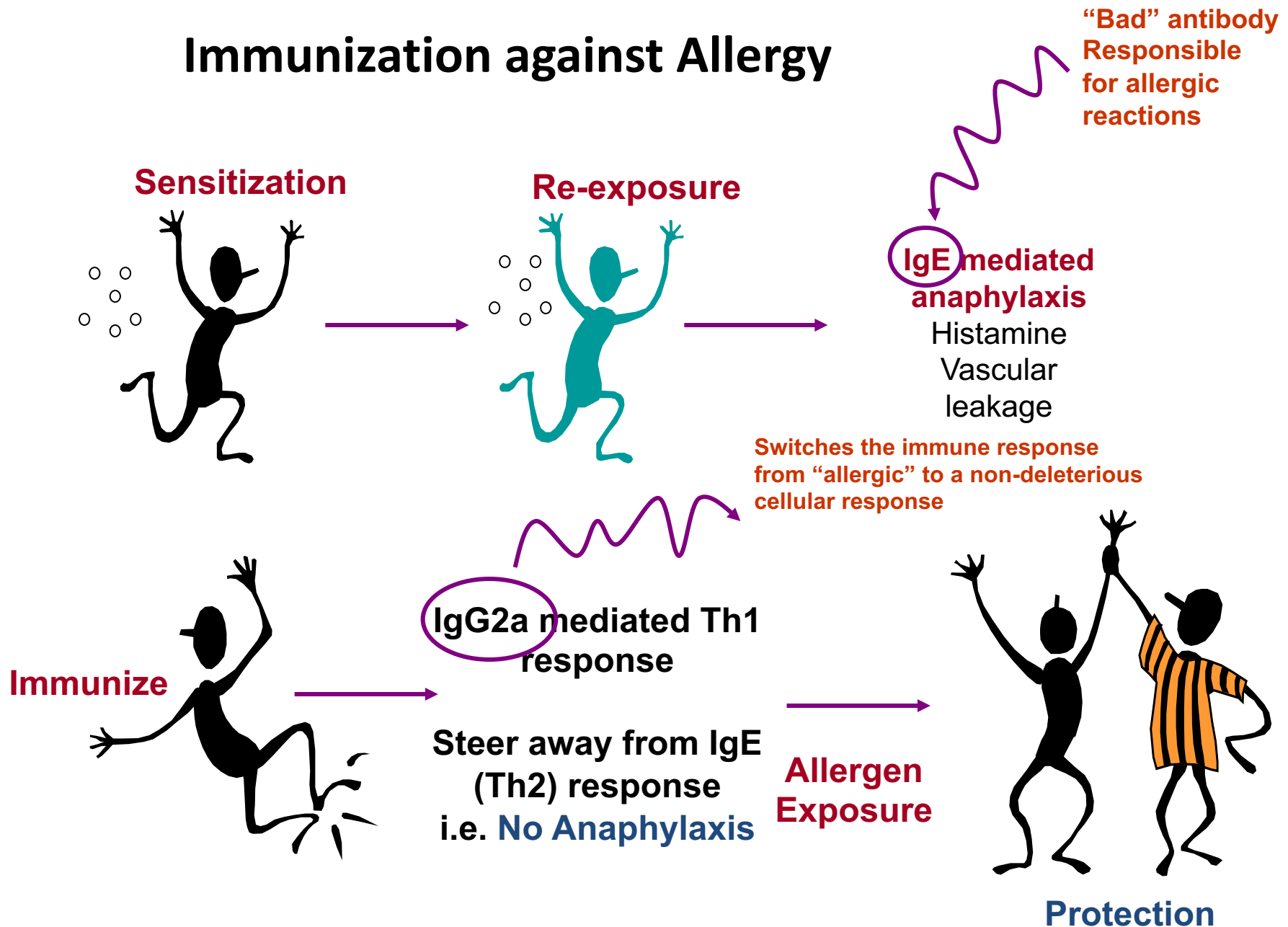


Serum IgG titers in groups ($n = 10$) of BALB/c mice immunized with either PLG/CTAB-p55 gag DNA, blank PLG microparticles + DNA, or DNA alone at 1 μg dose. Antibody titers are geometric mean titers $\pm$ SE at 2 weeks post-second immunization (week 6) time point. The response from the PLG/CTAB-p55 DNA was significantly higher ($P < 0.05$) than other groups.

Immunization against food allergy

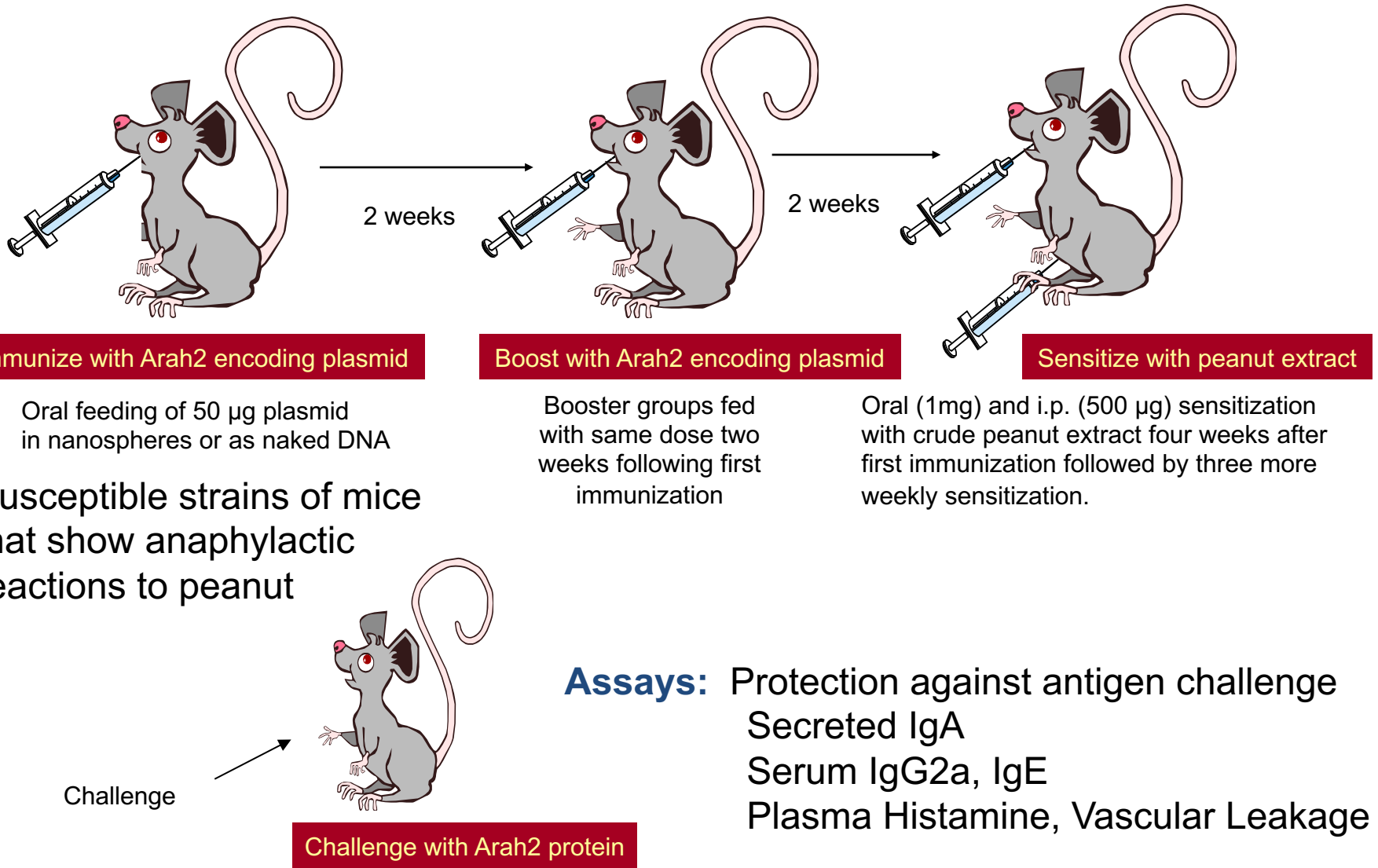
- **Peanut Allergy:**
 - Anaphylactic reactions mediated by Th2 cytokines (IgE and IgG1)
 - Potentially fatal or near-fatal especially in children
- **Allergen immunization:**
 - Potential preventive therapy if proven safe and effective
 - Protein administration can cause deleterious anaphylaxis
 - Need for novel immunization method that does not induce immunopathology
- **Oral immunization:**
 - High patient compliancy, ease of administration for mass vaccination
 - Generation of mucosal immunity against infectious agent

Immunization against Allergy

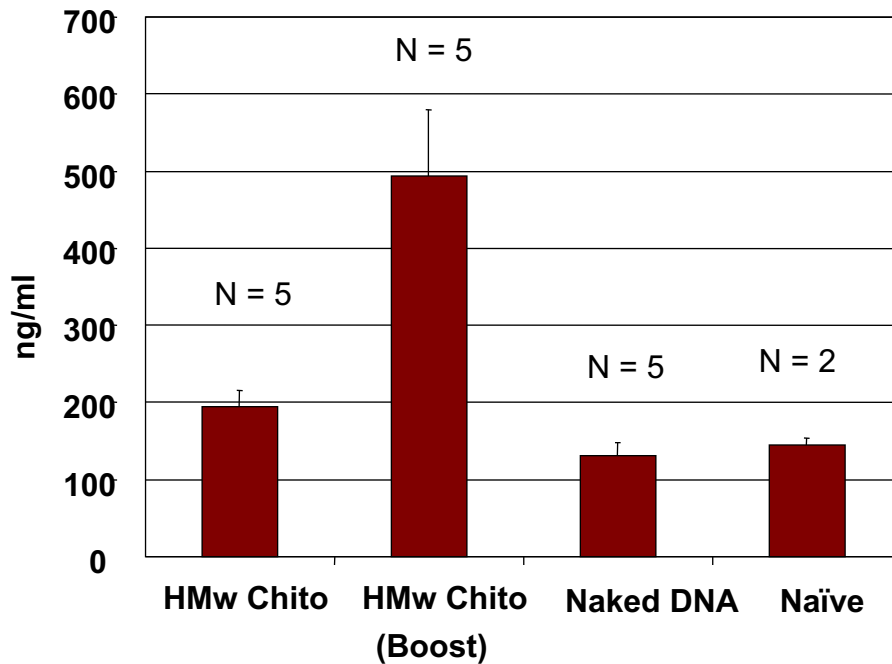


Oral DNA Delivery with Chitosan Nanoparticles

(Roy et al., Nature Medicine, April 1999)

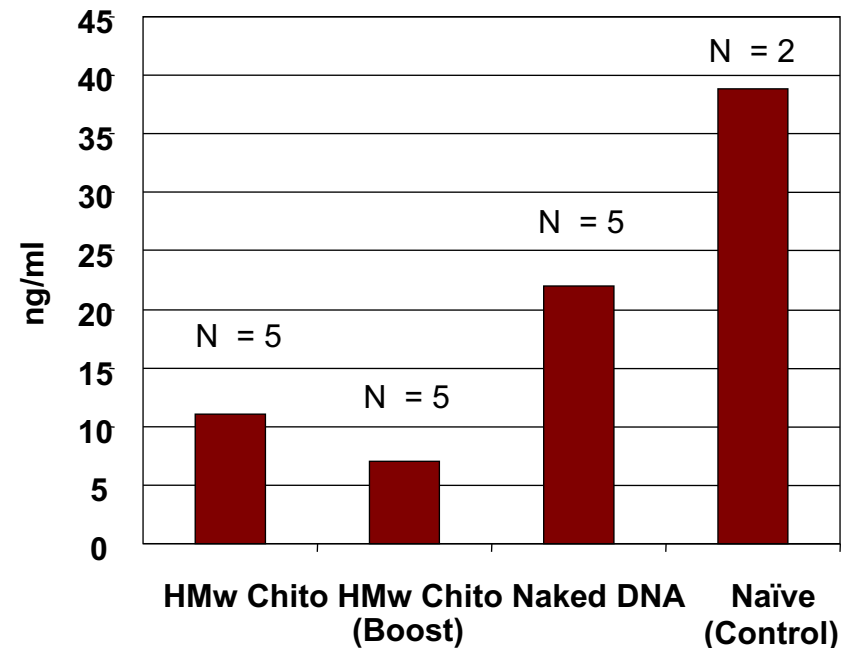


Chitosan-based oral DNA immunization switches the immune response from Th2 to Th1



Serum anti-Arah2 IgG2a response
four weeks after first immunization
(before sensitization)

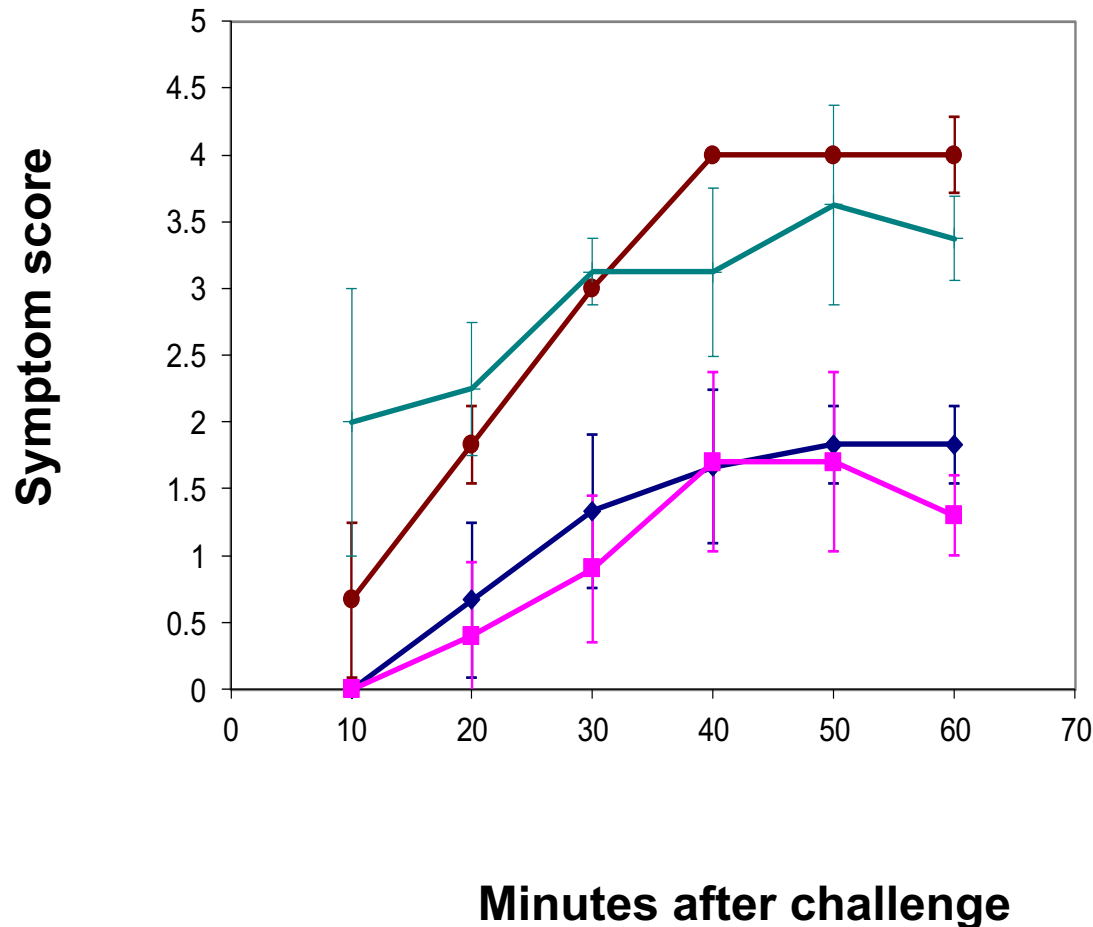
Therapeutic Th1 response



Increase in serum anti-Arah2 IgE
levels after three sensitizations

Anaphylactic Th2 response

Chitosan-based oral DNA immunization protects mice from Arah2 induced anaphylaxis



0: No sign of reaction

1: scratching and rubbing around the nose and head

2: decreased activity with an increasing respiratory rate, and/or puffing around the eyes

3: labored respirations, cyanosis around the mouth and tail

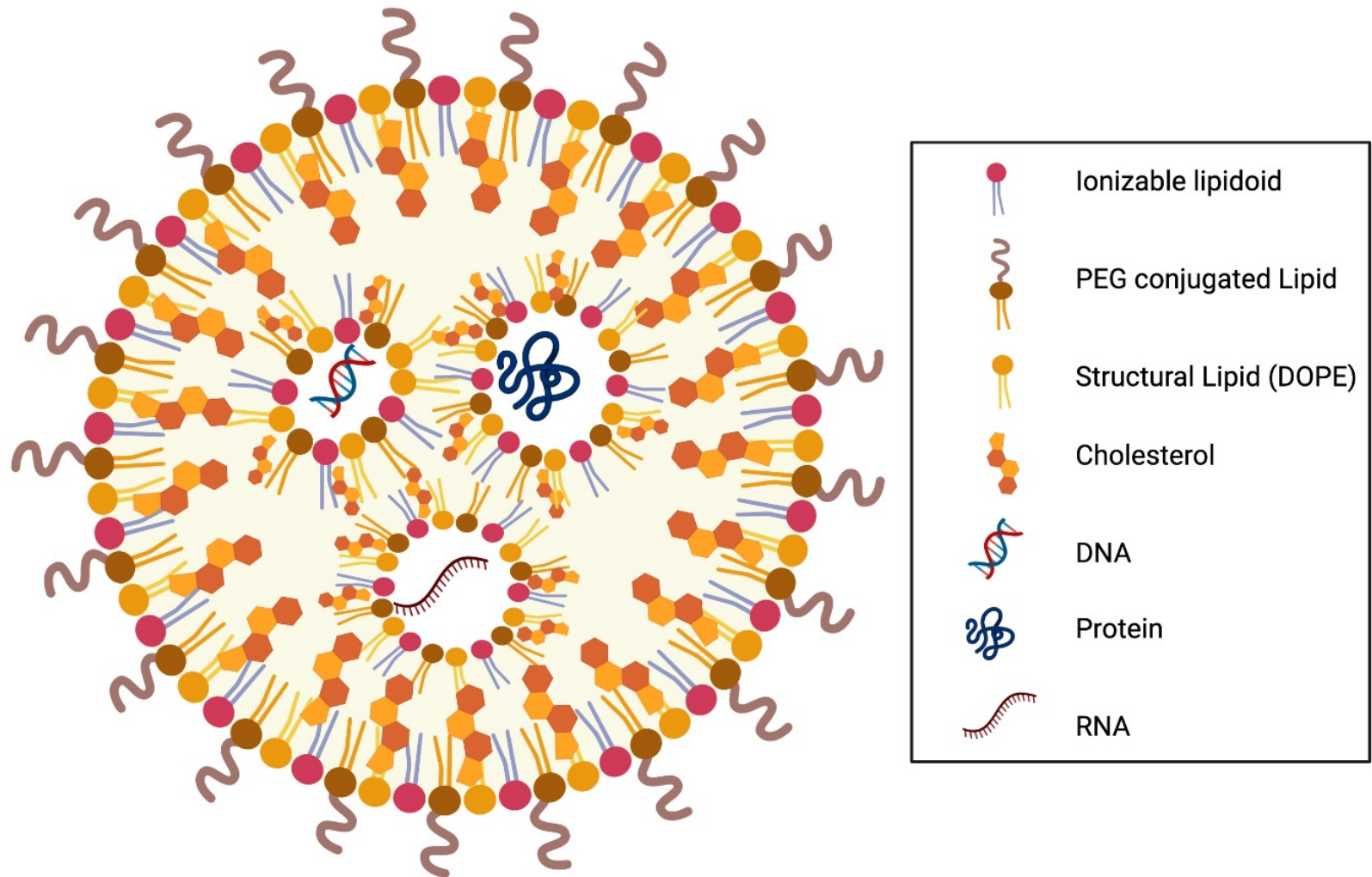
4: slight or no activity after prodding or tremors, convulsion

5: death

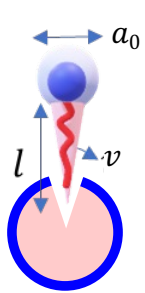
mRNA delivery

- Since gene delivery is challenging due to limitations of DNA diffusion in cell and translocation in the nucleus, mRNA delivery can overcome these barriers
- mRNA is a lot less stable than DNA
- Several discoveries (including 2023 Nobel prize in Medicine) related to mRNA modifications led to increased stability, less immunogenicity and efficient protein production!
- mRNA can be formulated with nanocarriers to increase their uptake and cytoplasmic expression and protect them from premature degradation
- A disadvantage is that the expression will be transient and would need repeated administration for sustained production
- Lipid nanoparticles have emerged as the delivery carrier of choice

Lipid nanoparticles

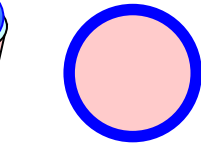


Self-Assembly of Lipids and Molecular Geometry

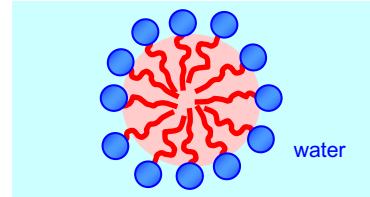


Critical Packing Parameter

$$CPP = \frac{v}{a_0 l} < 1/3$$



Spherical Micelles

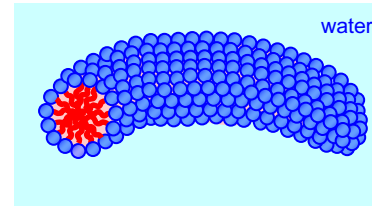


High positive curvature

$$\frac{1}{3} < CPP < 1/2$$

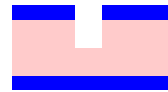


Cylindrical Micelles

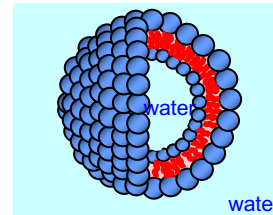


Intermediate curvature

$$\frac{1}{2} < CPP < 1$$

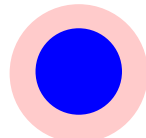
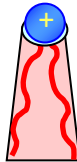


Bilayer Structures
(Liposomes etc.)

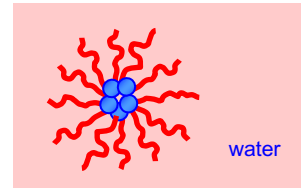


Low curvature

$$CPP > 1$$



Reverse Micelles



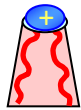
Negative curvature

v = tail volume

a_0 = head area

l = tail length

Structure of LNPs – Role of components

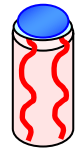


50% Ionizable Lipid (MC3)

$$CPP > 1$$



38.5% Cholesterol



10% Helper Lipid (DSPC)

$$\frac{1}{2} < CPP < 1$$



1.5% PEG-Lipid (DMG-PEG)

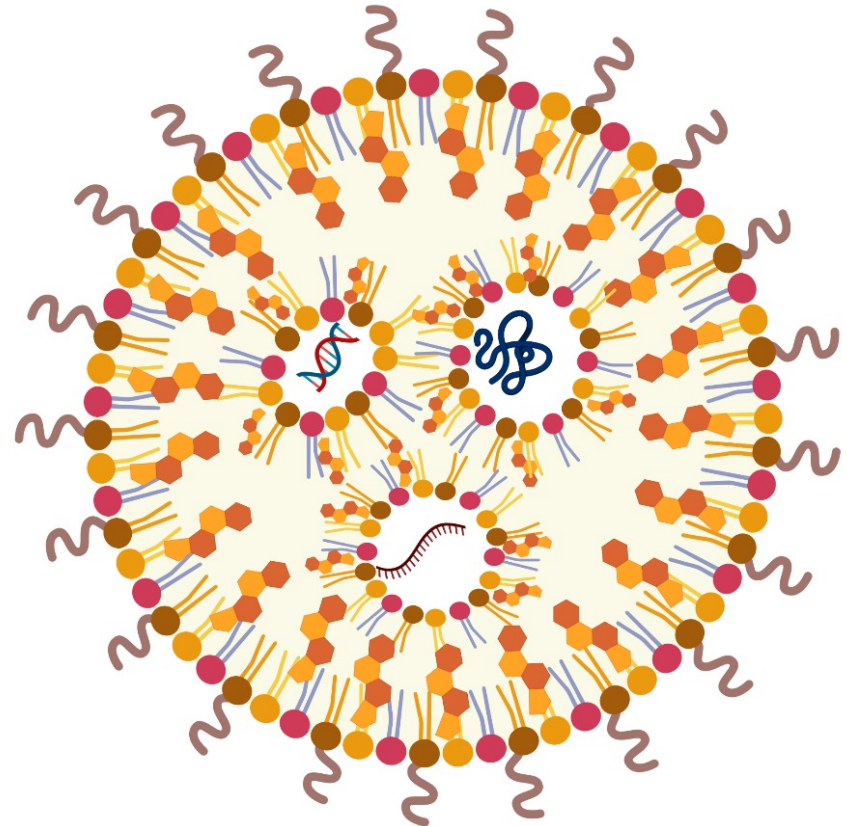
$$CPP < 1/3$$



mRNA

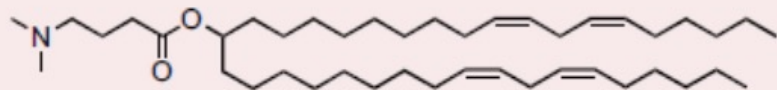


DNA

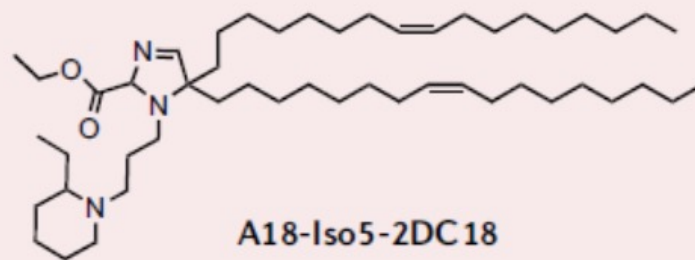


Ionizable lipids

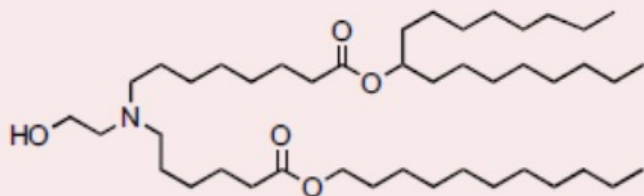
Ionizable lipid 



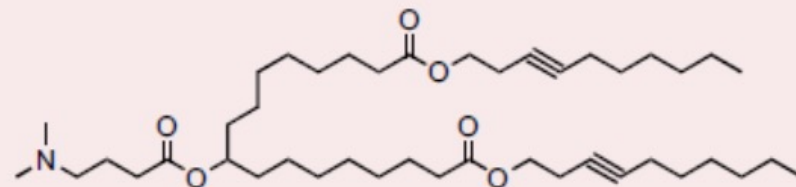
DLin-MC3-DMA (patisiran)



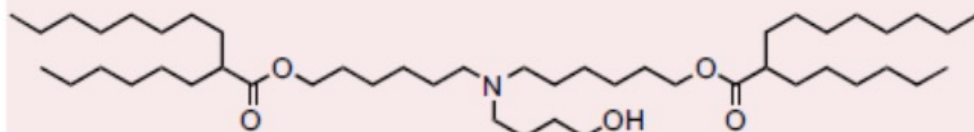
A18-Iso5-2DC18



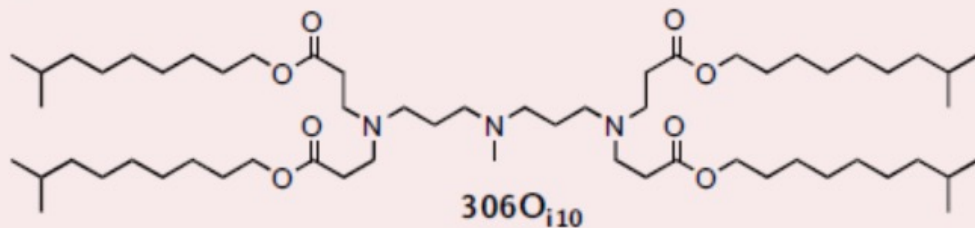
SM-102 (mRNA-1273)



A6



ALC-0315 (BNT162b2)



306O₁₀

Ionizable lipids

- Trapped in endosomes, in which the pH is lower than in the extracellular environment, ionizable lipids are protonated and, therefore, become positively charged, which may promote membrane destabilization and facilitate endosomal escape of the nanoparticles
- DLin- MC3- DMA; MC3 is a key delivery component of Onpattro (siRNA for polyneuropathies induced by hereditary transthyretin amyloidosis), the first United States Food and Drug Administration (FDA)- approved siRNA drug in 2018
- Incorporation of biodegradable lipids improves the tolerability of lipid nanoparticles, by allowing fast metabolism while retaining mRNA delivery efficacy. The biodegradability of lipids can be increased by introducing ester motifs, which shows good delivery efficacy and fast elimination from the liver and plasma in vivo

Table 2 | Representative clinical trials of lipid nanoparticle–mRNA therapeutics against infections, cancer and genetic disorders

Name	Disease	Encoded protein	Administration route	ClinicalTrials.gov identifier	Phase
<i>Infections</i>					
mRNA-1944	Chikungunya virus	Antibody against chikungunya virus	i.v.	NCT03829384	I
<i>Cancer</i>					
mRNA 2416	Solid tumours	OX40L	Intratumour	NCT03323398	II
mRNA-2752	Solid tumours	OX40L, IL-23 and IL-36 γ	Intratumour	NCT03739931	I
MEDI1191	Solid tumours	IL-12	Intratumour	NCT03946800	I
SAR441000	Solid tumours	IL-12 _{sc} , IL-15 _{sushi} , IFN α and GM-CSF	Intratumour	NCT03871348	I
<i>Genetic disorders</i>					
mRNA-3704	Methylmalonic acidemia	Methylmalonyl-CoA mutase	i.v.	NCT03810690	I/II
mRNA-3927	Propionic acidemia	Propionyl-CoA carboxylase	i.v.	NCT04159103	I/II
MRT5201	Ornithine transcarbamylase deficiency	Ornithine transcarbamylase	i.v.	NCT03767270	I/II
MRT5005	Cystic fibrosis	Cystic fibrosis transmembrane conductance regulator	Inhalation	NCT03375047	I/II
NTLA-2001	Transthyretin amyloidosis with polyneuropathy	CRISPR–Cas9 gene editing system	i.v.	NCT04601051	I

CoA, coenzyme A; CRISPR–Cas9, clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated protein 9; GM-CSF, granulocyte–macrophage colony-stimulating factor; IFN, interferon; IL, interleukin; i.v., intravenous.

